R V College of Engineering

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IV Semester B.E Fast-track/Supplementary Examinations - September/October-2025.
Electronics and Communication Engineering
Course : Transmission lines and electromagnetic fields-EC345AT

ime : 3 Hours **Maximum Marks : 100**

nstructions to the students

. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book nly.
. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full uestion from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

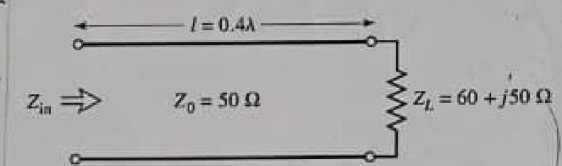
Part A

Question No	Question	M	CO	BT
1.1	A transmission line has a characteristic impedance of $50 + j0.01\ \Omega$ and is terminated in a load impedance of $73 - j42.5\ \Omega$. Calculate the magnitude of reflection coefficient and VSWR.	02	1	2
1.2	A distortion less line has $R = 6\,\Omega\,km^{-1}$, $G = 3.2\,\Omega^{-1}\,km^{-1}$ and $L = 8.847\,\mu\,H\,km^{-1}$. Find attenuation constant & capacitance of the line	02	1	3
1.3	Design a single-section quarter-wave matching transformer to match a 100Ω load to a 50 Ω transmission line at f0 = 5 GHz. Determine reflection coefficient for which the SWR ≤ 1.5	02	4	3
1.4	Given: $\vec{D} = 8\rho \sin\phi\,\hat{\rho} + 4\rho\cos\phi\,\hat{\phi}\ \mu C/m^2$. Find the volume charge density ρ_v at point $P(5, 30^\circ, -4)$	02	2	3
1.5	A spherical drop of water with charge 10μC has surface potential of 1000V. If 3 such drops combine to form a single drop, find the new surface potential.	02	1	2
1.6	Find the work done in carrying a $-10\,C$ charge from point $A(1, 2, -4)$ to point $B(3, -5, 6)$ in an electric field $$\vec{E} = x\,\hat{a}_x + z^2\,\hat{a}_y + 2yz\,\hat{a}_z \quad V/m.$$	02	1	3
1.7	Define Ampere's circuital law and derive its expression.	02	1	1
1.8	In free space magentic flux density is given as $10\sin(\pi y)\hat{a}_y + (4 + \cos(\pi x))\hat{a}_z Wb/m$. Find magnetic field intensity.	02	1	1
1.9	An EM field is said to be nonexistent or not Maxwellian if it fails to satisfy Maxwell's equations and the wave equations derived from them. Check whether the field $\vec{E} = 100\cos(\omega t)\hat{a}_x$ in free space is Maxwellian or not. Justify your answer.	02	4	4

Not maxwellian

1.10	If the dielectric constant (relative permittivity) of an MMIC FR-4 substrate is $\varepsilon_r = 4.4$, find the phase velocity and phase constant of an EM wave at an operating frequency of $f = 1\,\text{GHz}$.	02	2	3

Part B

Question No	Question	M	CO	BT
2a	Find the input impedance , reflection coeffcient and VSWR for a loaded lossy and lossless transmission line. And also explain special cases in loaded transmission line. Provide necessary expressions and diagrams.	10	1	1
2b	A lossy transmission line with a complex characteristic impedance of $Z_0 = 75 + j60\,\Omega$ is connected to a 200 Ω load. Given: - Attenuation constant $\alpha = 1.4\,\text{Np/m}$ - Phase constant $\beta = 2.6\,\text{rad/m}$ Find the input impedance for a line length of $\ell = 0.5\,\text{m}$.	06	4	2
3a	Use the Smith chart to find the following quantities for the transmission line circuit shown in the accompanying figure: (i) The SWR on the line. (ii) The reflection coefficient at the load. (iii) The input impedance of the line. (iv) The distance from the load to the first voltage minimum. (v) The distance from the load to the first voltage maximum. 	08	2	3
3b	If $\vec{D} = 2xyz\,\hat{a}_x + 3xy^2\,\hat{a}_y + 4xyz\,\hat{a}_z\ \text{C/m}^2$, find ρ_v at point $(1, 1, 1)$.	04	4	3
3c	State Gauss's law for electrostatics with necessary expressions.	04	2	1
	OR			
4a	A circular disk of radius a is uniformly charged with $\rho_s\ \text{C/m}^2$. The disk lies on the $z = 0$ plane with its axis along the z-axis. (i) Show that at point $(0, 0, h)$ $$\vec{E} = \frac{\rho_s}{2\varepsilon_0}\left\{1 - \frac{h}{[h^2 + a^2]^{1/2}}\right\}\hat{a}_z$$ (ii) From this, derive the $\vec{E}$ field due to an infinite sheet of charge on the $z = 0$ plane. (iii) If $a \ll h$, show that $\vec{E}$ is similar to the field due to a point charge.	08	2	2
b	Electric flux density vector, $\vec{D} = 2xyz\,\vec{a_x} + 3xy^2\,\vec{a_y} + 4xyz\,\vec{a_z}\ C/m^2$. Find the Volume charge density, ρ_v, at the point (1, 1, 1).	04	3	2

~~4c~~	Explain briefly about □Single-Stub Tuner (Matching) with necessary diagrams.	04	3	1
~~5a~~	Given a surface charge density of 8 nC/m^2 on the plane $x = 2$, a line charge density of 30 nC/m is positioned at $x = 1, y = 2$, and a 1 μC point charge at $P(-1, -1, 2)$. Find V_{AB} for points $A(3, 4, 0)$ and $B(4, 0, 1)$.	08	1	5
~~5b~~	Region 1 ($z < 0$) contains a dielectric for which $\varepsilon_{r1} = 2.5$, while region 2 ($z > 0$) is characterized by $\varepsilon_{r2} = 4.0$. Let field $\vec{E}_1 = -30\hat{a}_x + 50\hat{a}_y + 70\hat{a}_z$ V/m find: (i) $\vec{D}_2$ (ii) The angle between $\vec{E}_2$ and the normal to the surface. (iii) The energy density in each region.	08	2	5
	OR			
~~6a~~	Conducting cylinders at $\rho = 2$ cm and $\rho = 8$ cm in free space are held at potentials of 60 mV and -30 mV, respectively. Using Laplace equation, (i) Derive and determine V and $\vec{E}$ in the region between the shells. (ii) Determine the maximum $\vec{E}$ if the region has $\varepsilon_r = 2.5$. (iii) Determine the surface where $V = 30$ mV. (iv) The capacitance of the capacitor.	08	2	4
6b	A volume charge distribution with spherical symmetry is $\rho_V = \begin{cases} \rho_0, & 0 \le r \le R \\ 0, & r \ge R \end{cases}$ (i) Find $\vec{E}$ and V for $r > a$ (ii) Find $\vec{E}$ and V for $r < a$ (iii) Find the total charge.	08	1	5
~~7a~~	Line $x = 0, y = 0, 0 \le z \le 10$ m carries current 2 A along $\hat{a}_z$. Calculate $\vec{H}$ at points (i) $(5, 0, 0)$ (ii) $(5, 15, 0)$ (iii) $(5, 5, 0)$ (iv) $(5, -15, 0)$	08	2	2
7b	For a current distribution in free space, $\vec{A} = (2x^2y + yz)\,\hat{a}_x + (xy^2 - xz^3)\,\hat{a}_y - (6xyz - 2x^2y^2)\,\hat{a}_z$ Wb/m. (i) Calculate $\vec{B}$. (ii) Find the magnetic flux through a loop described by $x = 1, 0 < y < 2, 0 < z < 2$.	08	2	3

	(iii) Show that $\nabla \cdot \vec{A} = 0$ and $\nabla \cdot \vec{B} = 0$.			
	OR			
~~8a~~	Derive the boundary conditions for magnetic media and explain with suitable diagram.	08	2	2
~~8b~~	Two current elements $I_1\,dl_1 = 4 \times 10^{-5}\hat{a}_x$ A.m at $(0,0,0)$ and $I_2\,dl_2 = 6 \times 10^{-5}\hat{a}_y$ A.m at $(0,0,1)$ are in free space. Find $\vec{H}$ at $(3,1,-2)$.	08	3	3
9a	Using Maxwell's equations and Lorenz condition for potentials, arrive at expressions for scalar electric and vector magnetic potentials for time-varying conditions. Also, briefly explain why are they called retarded potentials?	10	4	1
9b	An $\vec{H}$ field in free space is given as $\vec{H}(x,t) = 10\cos(10^8 t - \beta x)\hat{a}_y$ A/m. Find (i) β (ii) λ (iii) $\vec{E}(x,t)$ at $P(0.1, 0.2, 0.3)$ at $t = 1$ ns.	06	4	3
	OR			
10a	The electric field component of a uniform plane wave traveling in seawater ($\sigma = 4\,\text{S/m}$, $\varepsilon = 81\varepsilon_0$, $\mu = \mu_0$) is $\vec{E} = 8e^{-0.1z}\cos(\omega t - 0.3z)\,\hat{a}_x$ V/m (i) Determine the average power density. (ii) Find the depth at which the power density is reduced by 20 dB.	08	1	4
10b	The electric field intensity in the region $0 < x < 5$ m, $0 < y < \pi/12$ m, $0 < z < 6$ m in free space is given by $\vec{E} = C\sin(12y)\sin(az)\cos 2 \times 10^{10} t\,\hat{a}_x$ V/m. Beginning with the $\nabla \times \vec{E}$ relationship, use Maxwell's equations to find a numerical value for a, if it is known that a is greater than zero.	08	4	4

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